

Portal Radio Bluetooth Speaker

Rainy Cobra

VIEW IN BROWSER

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Summary

Remixed Portal Radio with functional knob as both a power switch and volume with usb-c charging

Costumes & Accessories > Props

Tags: functional led replica prop portal speaker bluetooth radio aperture aperturescience bluetoothspeaker portalradio

Remix of Aibot's Portal Radio on MMF, with some quality of life features and changes. I added a rear hole for a usb-c port in the base, and some more space under the ark led for cable management. The antenna holder has a bottom to it now, making it way easier to secure the rod inside. And the knob actually works, both turning on the radio and being the volume. This was more made for show than to be functional and sound good, so don't expect audiophile quality from it.

Body, base, front panel, and electronics mounts were printed in PLA

Antenna rod mount, knob assembly, grills, and front screen were resin printed, with the ark and screen diffuser being printed in clear resin.

Shell is mounted with small bolts through the bottom with a nut that rests inside the top shell. Heat set inserts can be used for mounting the electronic mounts and ark/front more securely.

Feel free to use a proper battery setup, I just had to finish this before a certain date so I just jankily slapped an existing power bank inside and am pulling power from a micro-usb.

I wired the power switch on the amp to also be the power switch for everything, so when you turn the knob to its “on” state, the whole radio turns on.

Parts I used

Power Switch/Amp - https://www.amazon.com/dp/B0CFY1BPMS?ref=ppx_yo2ov_dt_b_fed_asin_title

Speakers - https://www.amazon.com/dp/B0826Y1BMY?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

Battery - https://www.amazon.com/dp/B08QHG7BTB?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

Antenna (happens to be the same length as in-game) - https://www.amazon.com/dp/B08L7WWHHK?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

12v step up - https://www.amazon.com/dp/B0BG465ZRK?ref=ppx_yo2ov_dt_b_fed_asin_title

Bluetooth module - https://www.amazon.com/dp/B08D94PGCG?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

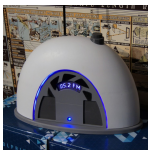
rear usb-c port - https://www.amazon.com/dp/B08BYQKJL6?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

micro usb 5v output - https://www.amazon.com/gp/product/B07B5ZDLJY/ref=ppx_yo_dt_b_search_asin_title?ie=UTF8&psc=1

Led strips (I cut the strips out of the silicone housing they were in to fit in the ark) - https://www.adafruit.com/product/3861

Led strip other option - https://www.ebay.com/itm/274148208415?var=574271725717

This remix is based on



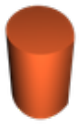
3D Printable Portal radio

by Alex I

Model files



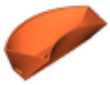
front-panel.stl



screen-8-top.stl



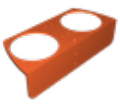
screen-diffuser.stl



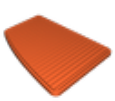
ledholder-screen.stl



knob-mount.stl



electronics-mount.stl



grill-left.stl



led-ark-diffuser.stl



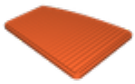
knob.stl



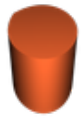
front-logo-stencil-1.stl



led-ring.stl



grill-right.stl



screen-8-bot.stl



base-fix.stl



led-ark.stl



shell.stl



screen-main.stl



antenne-holder.stl

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